

Gerrymandering Metrics

```
library(tidyverse)
```

Read the RAW precinct-level data directly

(make sure this path matches your repo: usually “data/
g24_sov_by_g24_svprec.csv”)

```
g24_raw <- readr::read_csv("data/g24_sov_by_g24_svprec.csv")
```

Do minimal cleaning here so this document is self-contained

```
g24_clean <- g24_raw |> # keep only 2024 general election, precinct-level rows filter(ELECTION == "g24", GEO_TYPE == "svprec") |> # drop statewide summary rows where district is 0 filter(CDDIST != 0) |> # make vote columns numeric and replace NAs with 0 mutate( CNGDEM01 = replace_na(as.numeric(CNGDEM01), 0), CNGDEM02 = replace_na(as.numeric(CNGDEM02), 0), CNGREP01 = replace_na(as.numeric(CNGREP01), 0), CNGREP02 = replace_na(as.numeric(CNGREP02), 0) )
```

Build district-level results for the 2024 congressional map

```
cng_2024 <- g24_clean |> group_by(CDDIST) |> summarise( dem_votes = sum(CNGDEM01 + CNGDEM02, na.rm = TRUE), rep_votes = sum(CNGREP01 + CNGREP02, na.rm = TRUE), total_votes = dem_votes + rep_votes, dem_share = dem_votes / (dem_votes + rep_votes), rep_share = rep_votes / (dem_votes + rep_votes), .groups = "drop" )
```

Function: mean–median score (Democratic vote share)

```
mean_median_score <- function(dem_share) { mean(dem_share) - median(dem_share) }
```

Function: efficiency gap

EG = (wasted_D - wasted_R) / total_votes

positive EG => Republican advantage

```
efficiency_gap <- function(dem_votes, rep_votes) { total <- dem_votes + rep_votes half_plus1 <- floor(total / 2) + 1
```

```
dem_wasted <- ifelse( dem_votes > rep_votes, dem_votes - half_plus1, # wins dem_votes # losses )
```

```
rep_wasted <- ifelse( rep_votes > dem_votes, rep_votes - half_plus1, # wins rep_votes # losses )
```

```
tibble( total_dem_wasted = sum(dem_wasted), total_rep_wasted = sum(rep_wasted),  
  efficiency_gap = (total_dem_wasted - total_rep_wasted) / sum(total) ) }
```